Answer Key - Mathematics - K-12 Level Assessment 12 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) \$10.00; B) \$10.50; C) \$13.50; D) \$16.00
   Note: The correct answer is 10.00$becausethecostperballis$0.10 (calculated by dividing 4.00$by 40 balls), so for 100 balls, the total cost is$0.10 multiplied by 100, which equals \$10.00.

2. **Answer:** B
   Options: A) 18; B) 24; C) 25; D) 29
   Note: The median is the middle value of a data set when it is arranged in ascending order, and in the sorted set 13, 18, 23, 25, 29, 32, the median is calculated as the average of the two middle numbers (25 and 23), which is 24.

3. **Answer:** A
   Options: A) 18 x+15; B) 18 x+3; C) 7 x+8; D) 2 x+8
   Note: The expression 5(4 x + 3) - 2 x simplifies to 20 x + 15 - 2 x, which combines to give 18 x + 15, demonstrating the distributive property and combining like terms.

4. **Answer:** C
   Options: A) 156; B) 110; C) 94; D) 48
   Note: The expression simplifies to 2(3(16 + 1)) - 8, which calculates to 2(3(17)) - 8, resulting in 102 - 8, and ultimately equals 94.

5. **Answer:** C
   Options: A) 0.25 1.6 1.0; B) 1.0 0.25 1.6; C) 0.25 1.0 1.6; D) 1.6 1.0 0.25
   Note: The correct answer is 0.25, 1.0, 1.6 because these numbers are arranged in ascending order based on their decimal values, where 0.25 is less than 1.0, and 1.0 is less than 1.6.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: The percentage of variation in y explained by x is determined by the square of the correlation coefficient, so a correlation of 0.6 explains 36% of the variation, while 0.3 explains only 9%, making the first not twice the second.

7. **Answer:** True
   Options: A) True; B) False
   Note: Tara's rate of developing film allows her to complete 8 rolls in 1 hour and 12 minutes, which is equivalent to 72 minutes, confirming the statement as true since 72 minutes divided by 8 rolls equals 9 minutes per roll.

8. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because after one half-life of 8 years, half of the substance remains, and after an additional 4.68 years (or approximately 0.585 of another half-life), about two-thirds of the original substance will have decayed.

9. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because the time it takes to wash a car at a car wash typically ranges from 5 to 10 minutes, not 16 minutes.

10. **Answer:** True
    Options: A) True; B) False
    Note: Joe finished the race at 3:58 p.m., which is indeed 4 minutes earlier than Mike's finish time of 4:02 p.m., confirming the statement is true.

Long Form Response

11. **Answer:** Since $\mathbf{v}_n$ is always a projection onto $\mathbf{v}_0$,

$$\mathbf{v}_n = a_n \mathbf{v}_0$$

for some constant a_n. Similarly,

$$\mathbf{w}_n = b_n \mathbf{w}_0$$

for some constant b_n. [asy] unitsize(1.5 cm); pair[] V, W; V[0] = (1,3); W[0] = (4,0); V[1] = (W[0] + reflect((0,0),V[0])*(W[0]))/2; W[1] = (V[1] + reflect((0,0),W[0])*(V[1]))/2; V[2] = (W[1] + reflect((0,0),V[0])*(W[1]))/2; W[2] = (V[2] + reflect((0,0),W[0])*(V[2]))/2; V[3] = (W[2] + reflect((0,0),V[0])*(W[2]))/2; W[3] = (V[3] + reflect((0,0),W[0])*(V[3]))/2; draw((-1,0)–(5,0)); draw((0,-1)–(0,4)); draw((0,0)–V[0],red,Arrow(6)); draw((0,0)–W[0],red,Arrow(6)); draw((0,0)–V[1],red,Arrow(6)); draw((0,0)–W[1],red,Arrow(6)); draw((0,0)–V[2],red,Arrow(6)); draw((0,0)–W[2],red,Arrow(6)); draw(W[0]–V[1]–W[1]–V[2]–W[2],dashed); label("$\mathbf{v}_0$", V[0], NE); label("$\mathbf{v}_1$", V[1], NW); label("$\mathbf{v}_2$", V[2], NW); label("$\mathbf{w}_0$", W[0], S); label("$\mathbf{w}_1$", W[1], S); label("$\mathbf{w}_2$", W[2], S); [/asy] Then **v**_n &= *proj*_{**v**_0} **w**_{n - 1}
&= $\frac{}{\{\mathbf{w}\_\{n-1\}\cdot\mathbf{v}\_0\}\{\|\mathbf{v}\_0\|^2\}\mathbf{v}\_0\&=\frac{\{}{b}\_\{n-1\}\cdot\mathbf{w}\_0\cdot\mathbf{v}\_0\}\{\|\mathbf{v}\_0\|^2\}\mathbf{v}\_0\&=\frac{\{}{b}\_\{n-1\}\left(4\right)0\cdot\left(1\right)3\}\{\left\|\left(1\right)3\right\|^2\}\mathbf{v}\_0\&=\frac{2}{5}b\_\{n-1\}\mathbf{v}\_}$
$a_n = \frac{2}{5}b_{n-1}$. Similarly, **w**_n &= *proj*_{**w**_0} **v**_n
&= $\frac{}{\{\mathbf{v}\_n\cdot\mathbf{w}\_0\}\{\|\mathbf{w}\_0\|^2\}\mathbf{w}\_0\&=\frac{\{}{a}\_n\cdot\mathbf{v}\_0\cdot\mathbf{w}\_0\}\{\|\mathbf{v}\_0\|^2\}\mathbf{w}\_0\&=\frac{\{}{a}\_n\left(1\right)3\cdot\left(4\right)0\}\{\left\|\left(4\right)0\right\|^2\}\mathbf{w}\_0\&=\frac{1}{4}a\_n\mathbf{w}\_0.}$ Thus,
$b_n = \frac{1}{4}a_n$. Since $b_0 = 1$, $a_1 = \frac{2}{5}$. Also, for $n \geq 2$,

$$a_n = \frac{2}{5}b_{n-1} = \frac{2}{5} \cdot \frac{1}{4}a_{n-1} = \frac{1}{10}a_{n-1}.$$

Thus, (a_n) is a geometric sequence with first term $\frac{2}{5}$ and common ratio $\frac{1}{10}$, so **v**_1 + **v**_2 + **v**_3 + &= $\frac{2}{5}$ $\mathbf{v_0}$ + $\frac{2}{5} \cdot \frac{1}{10} \cdot$ **v**_0 + $\frac{2}{5} \cdot$ ($\frac{1}{10}$)^2 $\cdot$ **v**_0 +
&= $\frac{2/5}{1-1/10}$ **v**_0 = $\frac{4}{9}$ **v**_0 = {(4) /9

4/3}.
Note: correct because the projections of the vectors maintain a consistent relationship with the original vectors $\mathbf{v}_0$ and $\mathbf{w}_0$, leading to the conclusion that $\mathbf{v}_n$ can be expressed as a scalar multiple of $\mathbf{v}_0$ and $\mathbf{w}_n$ as a scalar multiple of $\mathbf{w}_0$, thus revealing a pattern in the sequence sums.

Answer: To calculate the total cost of purchasing 4 magazines at $2.99 each and 3 books at 6.99$ each, we first find the cost of the magazines by multiplying the number of magazines by their price: 4 x 2.99 =11.96. Next, we calculate the cost of the books by multiplying the number of books by their price: 3 x 6.99 =20.97. Adding these two amounts together gives us the total cost: 11.96+20.97 = $32.93. Budgeting is important in this scenario because it helps ensure that we can afford the total cost and avoid overspending.
Note: correct because it accurately applies multiplication to determine the individual costs of magazines and books, and then correctly adds those amounts to find the total cost, illustrating effective budgeting by assessing affordability for the purchases.

End of Answer Key